

Probability for Machine Learning

My revision notes, built for understanding not memorizing

Why bother learning this at all

Every ML model is basically a guessing machine. It looks at data and makes a guess about what will happen next. Probability is the language that lets you say how confident that guess is. Without it, a model would just say yes or no with no idea of how sure it actually is.

Two very common ML algorithms are built directly on probability: Naive Bayes and Logistic Regression. If the basics in this note make sense, those two algorithms will feel like common sense instead of magic.

1. The five basic building blocks

Before you can calculate any probability, you need a shared vocabulary. These five words show up in every single problem you will ever see, so get comfortable with them first.

Random Experiment

Any action where you cannot say for sure what the result will be before you do it. The key word is unpredictable. If you already know the answer, it is not a random experiment.

- **Example:** tossing a coin, rolling a die, checking tomorrow's weather, or picking a card from a deck.

Trial

One single time you actually run the experiment.

- **Example:** tossing the coin once is one trial. Toss it 10 times and you have run 10 trials.

Outcome

The actual result you got from one trial. Just one single result, nothing more.

- **Example:** you tossed the coin and got Head. Head is the outcome.

Sample Space

The full list of every single thing that could possibly happen. This matters because every probability calculation needs to know the complete universe of possibilities first. It is basically the bottom half of every probability fraction.

- **Example:** coin toss sample space is {Head, Tail}. Rolling a die gives {1, 2, 3, 4, 5, 6}.

Event

A group of outcomes you actually care about, pulled out of the sample space. You almost never care about one exact outcome, you care about a pattern or a condition.

- **Example:** when rolling a die, "getting a number less than 3" is the event {1, 2}. "Getting an even number" is the event {2, 4, 6}.

2. Types of events

Once you know what an event is, you need to describe how events behave and relate to each other. This is where a lot of confusion usually starts, so pay close attention to the difference between independent and mutually exclusive further down.

Simple vs Compound

- Simple event: only one outcome in it. Example, rolling exactly a 3.
- Compound event: two or more outcomes grouped together. Example, rolling an odd number (1, 3, or 5).

Independent Events

Knowing that one event happened tells you nothing new about the other. They do not affect each other at all.

- **Example:** flipping a coin and rolling a die at the same time. The coin landing on Heads has zero effect on what the die shows.
- **Example:** drawing a card, putting it back in the deck, shuffling, then drawing again. Since the card went back, the second draw does not care what the first draw was.

Dependent Events

The first event changes the odds for the second event.

- **Example:** drawing two cards without putting the first one back. If you draw an Ace first, there are now only 3 Aces left out of 51 cards, so the chance for the second draw changed.

Mutually Exclusive Events

Two events that can never happen at the same time in one single trial. If one happens, the other one is completely blocked out.

- **Example:** rolling a die once, you cannot get an odd number and an even number on that same roll.

The trap: Independent is not the same as Mutually Exclusive

This confuses almost everyone the first time. It actually works the opposite way you would guess. If two events are mutually exclusive, they are actually a very extreme case of dependent, because one event happening completely destroys any chance of the other one happening. True independence means the events do not interact at all, while mutually exclusive events interact so strongly that they block each other completely.

Quick comparison table

Independent: $P(A \text{ and } B) = P(A) \times P(B)$ **Mutually Exclusive:** $P(A \text{ and } B) = 0$

If $P(A \text{ and } B)$ is zero, the two events cannot be independent unless one of them was impossible to begin with.

Exhaustive, Impossible and Certain events

- Exhaustive: a group of events that together cover the entire sample space, so at least one of them has to happen.
- Impossible event: probability is 0, it will never happen. Example, rolling 7 on a normal die.
- Certain event: probability is 1, it will always happen. Example, rolling a number less than 10 on a normal die.

3. What probability actually means

Probability is just a number between 0 and 1 that tells you how likely an event is. Closer to 0 means it almost never happens, closer to 1 means it almost always happens, and 0.5 means it is a genuine coin flip.

Theoretical Probability

This is what you would expect based on pure logic, before you ever run the experiment. It assumes every outcome is equally likely.

Formula

$P(\text{Event}) = \text{favorable outcomes} / \text{total possible outcomes}$

Example: chance of Heads on a fair coin is 1 out of 2.

Empirical Probability

This is what you actually observe after running real trials. You are not assuming anything, you are just counting what happened.

Formula

$$P(\text{Event}) = \text{number of times it happened} / \text{total trials run}$$

Example: you tossed a coin 200 times and got 80 Heads, so empirical probability of Heads is $80/200 = 0.4$.

Why do we need both

Theoretical probability tells you what should happen in an ideal world. Empirical probability tells you what is actually happening in the real world, which might reveal the coin is not fair at all. The important connection is this: if you keep increasing the number of trials, empirical probability slowly gets closer and closer to theoretical probability. This is why data and large sample sizes matter so much in machine learning, more data means your observed probabilities get closer to the true ones.

4. Random Variables

Here is the actual problem this concept solves. Outcomes like Head, Tail, or Red Marble are words, not numbers, and you cannot do math on words. You cannot add up Heads and Tails or find the average of Red and Blue. A random variable is simply a rule that turns every outcome into a real number, so that you can finally do math on it, like averages and spread.

Think of it as a translator sitting between the messy real world outcome and the clean number a computer or formula actually needs.

Discrete Random Variable

Can only take a countable, separate set of values, usually whole numbers. There are gaps between possible values, nothing in between.

- **Example:** coin toss mapped as Head = 1, Tail = 0. A die roll can only be 1, 2, 3, 4, 5 or 6, never 3.5.

Continuous Random Variable

Can take any value at all within a range, including decimals with endless precision.

- **Example:** a student's GPA can be 8.35, 8.351, 8.3512 and so on. Height, weight, and temperature all work the same way.

5. Probability Distribution, PMF and PDF

A probability distribution is just a full map of every possible value a random variable can take, next to how likely each value is. Once your outcomes are numbers (random variables), this is the natural next step, seeing the whole picture at once instead of one probability at a time.

Discrete example: single die roll

$P(X=1) = 1/6$, $P(X=2) = 1/6$, and so on all the way to $P(X=6) = 1/6$. Every value here can be listed one by one because there are only 6 of them.

Why PMF and PDF exist

Listing every value works fine for a die, but it completely breaks down for something like height, where there are infinite possible decimal values between 150cm and 151cm. You cannot write out infinite rows in a table. So instead of a list, we use a mathematical function, a formula that tells you the probability (or density) for any value you plug in.

- PMF, Probability Mass Function: used for discrete variables. Example, Binomial Distribution, Bernoulli Distribution.
- PDF, Probability Density Function: used for continuous variables. Example, Normal Distribution (the classic bell curve).

6. Mean or Expected Value

This answers one question: if you repeated this random experiment an enormous number of times, what number would the results average out to? It is a weighted average, where more likely values pull the average more strongly toward themselves.

Formula

$$E[X] = \text{sum of (each value } x \text{ times its probability } P(x))$$

Example 1: single die roll

$E[X] = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6) = 3.5$. Notice 3.5 is not even a number you can actually roll, but it is exactly where the long run average settles.

Example 2: a simple bet, to show why this matters in real decisions

Imagine a game where you win 10 dollars with probability 0.2, and lose 2 dollars with probability 0.8. Expected value = $10(0.2) + (-2)(0.8) = 2 - 1.6 = 0.4$ dollars. On average you gain 40 cents per game, so over many rounds this bet is actually worth taking, even though most single rounds end in a loss. This is exactly how expected value is used in real decision making, including how ML models choose between different actions.

7. Variance

The mean only tells you the center point, it says nothing about how wild or how tight the results actually are around that center. Variance measures the spread, how far values typically wander away from the mean.

Formula

$$\text{Var}(X) = E[(X - E[X])^2] \text{ which is the same as } \text{Var}(X) = E[X^2] - (E[X])^2$$

The second version is usually faster to actually calculate with.

Why this matters: two games with the same mean can feel completely different

Game A always pays exactly 5 dollars. Game B pays either 0 or 10 dollars with 50/50 chance. Both have the exact same mean of 5 dollars, but Game B is far riskier because the outcome swings wildly. Variance is the number that captures that difference. This is exactly why in ML you never trust the average alone, a low variance prediction is far more reliable than a high variance one, even if both have the same average.

8. Venn Diagrams

A Venn diagram is a picture of the sample space. It solves a simple problem: formulas about unions, overlaps, and complements are hard to picture in your head, so we draw them instead. A big rectangle represents everything possible (the whole sample space, probability 1). Circles inside it represent events, and where circles overlap represents outcomes shared by both events.

- **Example:** rolling a die, Event A = number is 4 or more, giving {4, 5, 6}, so $P(A) = 1/2$. The complement of A (everything not in A) is {1, 2, 3}, also $P = 1/2$.

For two overlapping events, the picture directly shows this classic formula:

Formula

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

You subtract the overlap once because simply adding $P(A)$ and $P(B)$ would count the overlapping part twice.

9. Contingency Tables

This is the same idea as a Venn diagram, just written as numbers in a grid instead of circles in a picture. It solves the problem of scale, you cannot draw a neat circle diagram for a real dataset with thousands of rows, but you can absolutely build a table of counts. Rows represent categories of one variable, columns represent categories of another, and each cell is a count of how many times that combination showed up.

Example: 100 students, subjects taken

- Only Biology: 40 students
- Only Math: 30 students
- Both Math and Biology: 10 students
- Neither subject: 20 students

This single table can instantly answer joint, marginal, and conditional questions, which is exactly why it becomes so important for the next three topics.

10. Joint Probability

This answers: what is the chance that two things happen together, at the same time. It matters because in the real world, almost nothing happens in total isolation, most events are tangled up with other variables, and you often need to know how two variables behave as a pair, not just separately.

Formula

$$P(\mathbf{A \text{ and } B}) = P(\mathbf{A \cap B})$$

Read as the probability of A intersect B, meaning both happen together.

Example: Titanic dataset

Using passenger class (Pclass) and survival status together, out of 891 total passengers, 80 were in first class and died. Joint probability of Pclass=1 and died = $80 / 891$. Every single combination of Pclass and Survived together forms the full joint distribution, and all of those combined probabilities add up to exactly 1.

11. Marginal Probability

This answers: what is the chance of just one variable, completely ignoring the other one. The name comes from where you find it, literally in the margins (the row totals and column totals) of a contingency table. You get it by adding up all the joint probabilities across the variable you want to ignore.

Formula

$$P(\mathbf{A}) = \text{sum of } P(\mathbf{A \text{ and } B}) \text{ across every possible value of } B$$

Example: Titanic dataset

Total first class passengers regardless of survival = 452 out of 891, so $P(\text{Pclass}=1) = 452/891$, roughly 0.51. This ignores survival completely, it only cares about class.

12. Conditional Probability

This is one of the most useful ideas in the entire topic. It answers: given that A has already happened, what is now the chance of B. The intuition is simple, you shrink your entire universe down to only the cases where A happened, throw away everything else, and then ask what fraction of that smaller world also has B.

Formula

$$P(B | A) = P(A \text{ and } B) / P(A)$$

Read as probability of B given A. The bottom $P(A)$ resizes the universe down to just A's world, the top counts how much of that smaller world also satisfies B.

Example: studying and passing

Say 60 percent of students study, and out of everyone, 50 percent both study and pass. Then $P(\text{pass} | \text{studied}) = 0.50 / 0.60 = \text{about } 0.83$. This tells you that among the students who studied, 83 percent of them passed, which is a much sharper and more useful number than just knowing the overall pass rate for the whole class.

Example: Titanic dataset

$P(\text{died} | \text{Pclass}=3)$, the chance of dying given you were in third class, works out to about 0.76. Compare this to the overall death rate of the whole ship, and you can immediately see third class passengers had it far worse than average. That is the entire point of conditional probability, it reveals patterns that a single overall average would completely hide.

In pandas, this is done with crosstab and a normalize argument. Setting normalize='columns' divides each column by its own total, giving you probability conditioned on the column category, and normalize='index' does the same thing but row wise.

13. Independent vs Dependent vs Mutually Exclusive, revisited with formulas

Now that you know joint and conditional probability, here is the same comparison from Section 2, but written properly with formulas.

Independent

$$P(B | A) = P(B) \text{ and } P(A \text{ and } B) = P(A) \times P(B)$$

Knowing A happened changes nothing about B.

Dependent

$$P(B | A) \text{ is not equal to } P(B)$$

Knowing A happened does change the odds for B.

Mutually Exclusive

$$P(A \text{ and } B) = 0 \text{ so } P(B | A) = 0$$

If A happened, B is now completely impossible.

Independence is what makes an algorithm called Naive Bayes possible. It assumes, sometimes wrongly but usefully, that all the features of your data are independent of each other, which massively simplifies the math.

14. Bayes Theorem

Sometimes you can only measure a conditional probability in one direction, but you actually need it in the other direction. For example, doctors can easily study $P(\text{positive test} \mid \text{actually has disease})$ by testing known patients, but what a patient really wants to know is the reverse, $P(\text{actually has disease} \mid \text{positive test})$. Bayes Theorem is the tool that flips a conditional probability around.

Formula

$$P(A \mid B) = [P(B \mid A) \times P(A)] / P(B)$$

$P(A)$ is called the prior, what you believed before new evidence. $P(B \mid A)$ is the likelihood, how well the evidence fits that belief. $P(B)$ is the evidence, the overall chance of seeing that evidence at all. $P(A \mid B)$ is the posterior, your updated belief after seeing the evidence.

Where the formula comes from

It is really just conditional probability rearranged. Start from $P(B \mid A) = P(A \text{ and } B) / P(A)$, rearrange it to $P(A \text{ and } B) = P(B \mid A) \times P(A)$, then plug that into the definition of $P(A \mid B) = P(A \text{ and } B) / P(B)$. No complicated proof needed, it falls straight out of the conditional probability formula you already know.

The classic example that makes this click: a medical test

A disease affects 1 percent of people. A test for it is 99 percent accurate, meaning it correctly detects sick people 99 percent of the time and correctly clears healthy people 99 percent of the time. You test positive. What is the actual chance you have the disease? Most people guess 99 percent. The real answer is far lower, around 50 percent, because the disease is so rare that even a very accurate test still produces plenty of false alarms among the huge pool of healthy people. This surprising gap is exactly why Bayes Theorem exists, gut feeling about probability is often wrong, and the math corrects it.

Why this matters for ML

Naive Bayes classifiers use exactly this formula to predict a category, like spam or not spam, by combining a prior belief with new evidence from the data (the words in the email), then producing an updated, corrected probability. This single formula is also the foundation of an entire branch of statistics called Bayesian statistics, which shows up again later when tuning model hyperparameters.

End of notes. Revisit Section 13 and the medical test example in Section 14 first if short on time, they are the two ideas that trip people up the most.